

Eco-Loop Building Agents

Project Statement ID: [paste from HirePro]

Domain: Building energy / Physical AI

Category: Software

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Objective: Safe autonomous building optimisation through EnergyPlus, MCP and deterministic AI control.

ed-loop Physical AI for buildings

Problem

Fixed HVAC schedules ignore occupancy & weather

Buildings waste energy and peak demand

Uncontrolled LLM setpoints are unsafe

Dashboards observe — they do not act

Solution

- EnergyPlus digital building sandbox
- OSS LLM (Ollama) + MCP tool bus
- SafetyShield gates every write
- Forward inject setpoints → next state

Flow: Observe → Propose → Validate → Execute → Measure → Correct

1. SafetyShield validates. EnergyPlus executes.

AgonyPlus · MCP · Ollama · SafetyShield

```
AgonyPlus → observations → MCP client → stdio → MCP server  
Ollama strategy / ECM  
Deterministic optimiser → SafetyShield  
Ag-SetP-Sch actuator → next state → self-correction
```

Runtime API actuator writes (not dashboard-only)

Spawn MCP process (`mcp\_pids\_differ: true`)

Full open-source LLM · safe deterministic fallback

Path A = savings proof · Path C = LLM/MCP agency

king closed loop

tor → next state

Path A approved writes
g-SetP-Sch` = True
llowing zone temperatures returned

Safety & agency

- 35°C proposal rejected
- Ollama-down → fallback (still gated)
- Path C self-correction logged

ashboard `DATA\_MODE=energyplus` shows measured KPIs (no silent mock)
demo: twinpilot.webyaar.in

A — comfort-zero savings

	BASELINE	ECO-LOOP	IMPROVEMENT
energy	13.85 kWh	13.16 kWh	
energy	421.51 kWh	416.00 kWh	
power	19.93 kW	19.64 kW	
ort violations	0 h	0 h	

e IDF · EPW · occupancy · period — controller is the only difference

C proves LLM strategy via MCP (agency); Path A is the % claim

ings without sacrificing occupied-zone comfort.

able PoC · honest limits

ility

ns on local / cloud VM
open-source local LLM
producible scripts
ean-clone verified
fe degraded mode

Limits & future

- Digital building only (no physical BMS yet)
- Scenario-specific savings
- Future: BACnet/Modbus
- Campus scale · tariffs · pilot